

Deployment Diagram

Project: LearnLens AI

1. Local-First Workstation Deployment Topology

```
deploymentDiagram
node "User Workstation [Windows 10/11, macOS, Linux]" as Workstation {

node "Web Browser [Google Chrome / Chromium]" as BrowserNode {
artifact "LearnLens React Studio\n[http://localhost:5173]" as WebStudio
artifact "Chrome DisplayMedia API\n[Tab / Window Screen Capture]" as CaptureAPI
artifact "LearnLens Companion Extension\n[Manifest V3 Side Panel]" as Ext
}

node "Python Runtime Environment [Python 3.10+]" as PyRuntime {
artifact "FastAPI Application Server\n[Uvicorn on 127.0.0.1:8000]" as APIServer
artifact "WebSocket Connection Manager\n[127.0.0.1:8000/ws]" as WSServer
artifact "Perceptual Deduplication & OCR Engine" as VisionEngine
artifact "python-pptx & ReportLab Generators" as DocGenerators
artifact "faster-whisper Engine [Optional GPU]" as WhisperEngine
}

node "Local Model Runtime [Optional]" as OllamaNode {
artifact "Ollama Server\n[http://localhost:11434]" as OllamaSvc
artifact "qwen2.5 / llama3 weights" as LLMWeights
artifact "qwen2.5-vl vision weights" as VLMWeights
}

node "Local Workstation Storage" as StorageNode {
database "SQLite Database\n[data/learnlens.db]" as SQLiteDB
folder "Session File Storage\n[data/sessions/<session_id>]" as SessionFolder {
file "Screenshots & Thumbnails" as Frames
file "Generated PPTX & PDFs" as Exports
file "Vector Inverted Index" as Vectors
}
}

WebStudio --> APIServer : HTTP / REST API
WebStudio --> WSServer : WebSocket Stream
CaptureAPI --> WebStudio : MediaStream
Ext --> APIServer : HTTP Queries

APIServer --> SQLiteDB : Read / Write Metadata
APIServer --> SessionFolder : Store Assets
APIServer --> OllamaSvc : Inference via HTTP
APIServer --> DocGenerators : Invoke Creation
WSServer --> VisionEngine : Pipe Frames
WSServer --> WhisperEngine : Pipe Audio
```